

Luiz Sérgio da Silva Filho

Full Stack Software Developer · AI (LLM) Solutions, Automation and API Integration

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Open to hybrid or remote roles

SUMMARY

Background in Information Technology, with hands-on experience in IT technical support, now focused on building full stack software that applies artificial intelligence in real, functional ways: structured document extraction, tool-calling agents, process automation and API integration. Running an independent business, I build complete systems on my own, with real databases, authentication, payments and integrations, including one system in production serving real customers and an AI agent that handles customer service over WhatsApp. I apply engineering practices such as multi-tenant data isolation with Row-Level Security, automated integration testing and cost control on AI calls.

PROFESSIONAL EXPERIENCE

Pedagogical Coordinator

Feb 2026 – present

Children's Coding & Robotics School · Barretos, SP

- Structured the pedagogical coordination from the course's launch: set up the classes, defined the curriculum by age group and trained the teaching staff, serving students from age 7 to 18.
- Age-based progressive curriculum: children's classes with robotics (Arduino, micro:bit) and block-based programming; teenage classes with Python as the starting language, advancing to Git, HTML, CSS, JavaScript, Tailwind CSS, daisyUI, React, Node.js, MongoDB, JWT and Swagger as a full stack development foundation.
- Teach Python to teenage students, covering programming logic, basic data structures and small practical projects.
- Faced with the limitations of the system the school was then using for pedagogical management and tracking, I owned the entire cycle of a system built to replace it, on my own: gathered requirements from my own pedagogical coordination routine, designed the data model and role-based permission model (manager, coordinator, teacher), planned the build in phases and validated each delivery against real school usage before shipping it to production.
- The AI layer woven into the pedagogical workflow was architected with automatic failover across multiple LLM providers for resilience, and with personal-data anonymization before any content reaches the AI.

Business Owner

Jun 2020 – present

Empório do Bebê · Barretos, SP

- Responsible for business operations: point-of-sale and inventory control system, online sales channel/digital catalog, and spreadsheet-based automation of administrative processes.
- In parallel, independently building a line of AI-powered software projects: complete systems with real databases, authentication, payments and integrations, including customer service automation, AI-based data extraction and one system in production with real customers.
- Multi-tenant data isolation with Row-Level Security in depth, real integration tests, cost engineering on AI calls and active prompt injection defense.

IT Support Analyst

Aug 2013 – Dec 2019

Sesi São Paulo · Limeira and Barretos, SP

- IT technical support for staff, teachers and students: helpdesk ticketing, administrative and educational systems, and classroom technology support.
- Technical mentoring of the educational robotics teams: supporting programming and the study of the robot's mission execution sequence to solve the proposed challenges, in competitions such as FLL (First Lego League) and OBR (Brazilian Robotics Olympiad).

INDEPENDENT TECHNICAL PROJECTS

Built independently, alongside my work as business owner at Empório do Bebê.

- Identity document verification system: extracts data from photographed documents using computer vision and checks the authenticity of the information (such as CPF checksum and date validity) directly on the server, never trusting the AI reading alone.
- AI sales agent serving customers over WhatsApp: checks real-time stock and prices and closes orders using real tool-calling against the database, with a safety guard that never accepts a price suggested by the AI without confirming it in the system.
- Multi-company SaaS platform: each customer uses the same system with data fully isolated from every other customer (multi-tenant architecture), protected by Row-Level Security at the database level and validated by an automated integration test suite.
- Inventory liquidation system in production: organizes products into batches and cards and runs auditable raffles using cryptographically generated random numbers, keeping the full financial history in an append-only ledger. Already serving real customers.
- Recurring practice across all AI projects: use multiple LLM providers (Claude and OpenAI) interchangeably, with per-call cost control, prompt caching to reduce latency, and active prompt injection defense.

EDUCATION

Postgraduate Degree in Higher Education Teaching 2015

Faculdade XV de Agosto

Bachelor's Degree in Information Technology 2008 – 2011

Faculdade Barretos

TECHNICAL SKILLS

Frontend & Languages: TypeScript, Python, React.js, Next.js

Backend & APIs: Node.js, NestJS, REST API, Webhooks

Data & Architecture: PostgreSQL, Prisma ORM, Drizzle ORM, Row-Level Security, Multi-tenant Architecture, Redis

Artificial Intelligence: Anthropic Claude API, OpenAI API, LLM Integration, Prompt Engineering, AI-Assisted Development

Infrastructure & Engineering Practices: Docker, CI/CD, Git, BullMQ, Automated Software Testing, Stripe API, Process Automation, n8n